

Projects: 625.714: Stochastic Differential Equations

I have used various routes for projects in other EP classes with varying degrees of success. For our 714 class we're going to try something that Carnegie Mellon has used for their upper-level statistics classes. What they used was similar to the reviews performed by the Neural Information Processing Systems (NIPS) conferences around categories of paper submissions and experts assembled for each category.

You will select a project category that you will endeavor to learn and become a 'near-expert' within. That is, you will pick a project to be performed by yourself on your own schedule during our summer-term. At each deliverable point, you'll post your project 'deliverables' to which other members of your 'near-expert' classmates who will be providing reviews. I'll continue to deliver comments to you privately and the grading will be off-line (i.e., not public). What I like about this way of handling a project is that you'll be able to share your project with other class members, get feedback from your peers, and have opportunities to learn and provide constructive criticisms to your peers' projects.

I think this will be fun for all of us. However, since this is my first attempt for this type of project, we'll undoubtedly have some issues to work-through along the way. Here's the initial guidelines:

- 1) I'm putting four categories below. I will ask each of you to submit to me your choice (ranked first category choice to fourth category choice). I will try to honor everyone's first choice, but I will also make the numbers of 'near-experts' for each category as close as possible to the same. By that I mean that we'll have $\frac{1}{4}$ of the class on each of our four topics
- 2) We have four public discussion points (week-8, week-9, week-12 and week-14). At each of these points, you'll turn in the project progress via the Blackboard discussion as well as at the to-be-graded location.
- 3) Each of you will be required to comment on the other submissions within the members of your category/cohort. So, if a cohort has five members, you'll need to realize that you will have four reviews from the rest of your members within your cohort and you will need to review four projects within your cohort. These don't need to be long, but should be constructive and supportive. Please be polite, respectful, and thoughtful. Abusive comments and words, and/or insults are not tolerated. I will be monitoring what is posted and *your feedback will be part of your project grade*. Think of this as helping your fellow classmates to better understand the course material.
- 4) Each of the four topic areas is also covered in the class material. So, if you are interested in more than one (ideally all four!), please do realize that this way to handle the projects is meant to help you learn more of the core material and that you will see all four of these topic areas within the class.
- 5) You can most certainly comment on project progress from contributors outside of your 'near-expertise'. This means that you will need give feedback to the rest of your members within your cohort at a minimum, but you are welcome to provide more feedback to any other project outside your cohort as well.

Topics will be selected from the following set of four broad areas. The topic choices, a ranking of your choices from one to four, is due at the end of the 5th Module. I fully realize that you'll still be learning as we reach that stage of the class, but these topics are sufficiently broad that learning about them can begin early and then will be further enhanced when we cover them, at least in part, during the class. I'm putting a paper that we also cover in the course modules that are associated with each topic.

Candidate Topic Areas:

1) Stochastic Control and forecasting

Those pursuing this area will become proficient in Kalman filters, as initially seen from recursive least squares and forecasting ideas for Stochastic Processes. Those of you with Bayesian statistics backgrounds can also view this as Kalman filters from that perspective. This has many broad application areas, and we cover this material at multiple times throughout the class. Most projects in this category will be using MATLAB, though the cohort is certainly allowed to use R or Python. A paper that we will cover in class and that gives some sense for what a 'near-expert' should master is the Young et al 1999 paper (material in Module 8).

Young, Peter C., Diego J. Pedregal, and Wlodek Tych. "Dynamic harmonic regression." *Journal of forecasting* 18, no. 6 (1999): 369-394.

2) Changes of Measure: Girsanov and Feynman-Kac

A foundational starting point for Black-Scholes is the ability to work backwards from the solution space. This relates to Feynman-Kac and measure changes (Girsanov). Projects in this area can be more pure math, applied math, or programming related. This area may be of special interest to those within the financial math program. A recent paper to help see the expertise gained from this project is here (material in Module 9):

Wang, Yichen, Grady Williams, Evangelos Theodorou, and Le Song. "Variational policy for guiding point processes." In *International Conference on Machine Learning*, pp. 3684-3693. 2017.

3) Diffusion Maps

As the data sets get larger and larger, an interesting question of analysis with Stochastic Processes is defined: can the highly dimensional data sets be inferred as sets of motion from dynamical systems with different time-courses? This approach works backwards from the data to try and infer insights, rather than from knowledge of the SDEs directly. Projects in this area could work with large datasets, be pure math, or could define new programming algorithms. A paper to give a sense for this project is (material in Module 10):

Banisch, Ralf, Zofia Trstanova, Andreas Bittracher, Stefan Klus, and Peter Koltai. "Diffusion maps tailored to arbitrary non-degenerate Ito processes." *arXiv preprint arXiv:1710.03484* (2017).

4) Optimal Stopping and Portfolio Optimization

With the right problem description, optimal stopping times and/or optimal portfolio returns for a reward function can be determined. Projects in this area may work with the Brownian Bridge and/or HJB to address how to optimize a reward within a Stochastic Process. Those interested in financial math may also find this project area of interest. It has broad applications to many aspects of decision theory. Two papers give a sense for this project (materials in Module 11 and Module 12):

Ekström, Erik, and Henrik Wanntorp. "Optimal stopping of a Brownian bridge." *Journal of Applied Probability* 46, no. 1 (2009): 170-180.

Dang, Duy-Minh, and Peter A. Forsyth. "Better than pre-commitment mean-variance portfolio allocation strategies: A semi-self-financing Hamilton–Jacobi–Bellman equation approach." *European Journal of Operational Research* 250, no. 3 (2016): 827-841.